E-Relationship Through a Value Lens--Implications for Organizational Capability.

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Abstract:

Firms that extend their e-commerce platforms into customer decision-making to build relationships need to understand the customer journey or build e-relationships through a lens that the customer uses to assess the value created and fulfilled. By contrasting the value cycles that customers and firms use, four different activities are identified for firms with shorter and longer value cycles. By enabling customers to support value creation, fulfilling the value created by enlisting partners and expanding services, and engaging the customers during value-in-use, three organizations discussed in the published literature appear to have developed e-relationship strategies to create and sustain value both before and during COVID-19. This analysis led us to develop three different insights: customer empowerment, collaborative competency, and intelligent digital platform for organizational capability. These insights were used to analyze two different organizations interviewed during the 2020period, as they faced many COVID-19 challenges, and assess the generalizability of these capabilities. We provide three recommendations for IS leaders for building e-relationships through a customer lens, by drawing on five case studies of companies to remain agile.

Keywords: Value creation, value fulfillment, e-relationship, intelligent digital platform, organizational capability, sustain value

Full Text:

INTRODUCTION

Businesses have used the primary and support activities of their business value chain to transform inputs into outputs and fulfill value propositions made to their customers, with the expectation that the products (including service products) will create value or utility for their customers. Such value creation through a proposition or promise is viewed more as a value-in-exchange between the firm and the customer, as both the firm and the customer will compare what they receive against their costs (Porter, 2010) and their benefits against sacrifices (Zeithaml, 1988). That is, the exchange will make both better off in some respect (Gronroos, 2008). Recognizing that market conditions continue to alter the way value is perceived by a customer, to stay competitive firms use feedback from customers to develop new value propositions and to fulfill these propositions by designing innovative products using their organizational resources. Let us refer to this as the business value cycle, which includes developing value propositions as a part of value creation, fulfilling value, and using feedback to assess value-in-use to repeat the next value cycle.

At the same time, customers have their own value cycle activities used to discover a need to solve a problem through the purchase of a product (i.e., value discovery), address this need by exploring options and purchasing a product (value fulfillment), and assess its ability to address the problem

(value-in-use assessment), before they decide to repeat this value cycle. Firms have used several demand side or customer-facing activities, such as marketing, sales, distribution, and service, to interact with customers as they fulfill their needs through their purchase journey (pre-purchase exploration, purchase, and post-purchase maintenance of the product). Advanced digitization, beginning with the Internet, has enabled firms to engage customers using e-commerce platforms. This has allowed firms to use more service interactions or exchanges with customers and develop value propositions as they learn more about their customers' purchase journeys (Bossert et al., 2014). These service interactions have included helping customers as they search for products that address their needs, allowing them to evaluate the firm's product offerings against others using the customers' decision criteria, giving them options as they seek to acquire the product, and helping them track and maintain the product while in use.

Ideally, firms would like to extend their interactions with customers to the discovery phase, when customers are determining what needs they must solve to address a problem. That is, firms want to co-create value with customers. In other words, with customer expectations changing rapidly due to technological advances, firms need to continue to align their value cycle activities with those of their customers and make their e-commerce platform become an e-relationship platform for creating and sustaining the value and building customer loyalty. Such a customer-centric and value-driven platform is the basic premise of service-dominant logic (Vargo & Lusch, 2014). The dynamic interaction between customers and firms, along with their many stakeholders, to create and sustain value along multiple value cycles means that firms must develop an organizational capability to continually transform themselves.

Building organizational capability, seen through the lens of customers, poses varied challenges for firms with longer product life cycles (e.g., durable product manufacturers or commodity firms). Organizations may not be able to change fast enough to expand products or become service driven, i.e., create shorter customer value cycles like professional service firms or customer service firms. Customer expectations change too fast, or digitization continues to alter the competitive landscape. The goal of this paper is to not rely on a firm-centric approach to compete but to use a customer-centric approach to develop an e-relationship strategy by interweaving customers and partners along the continuum of value creation and fulfillment. Such a strategy requires an integrated view of decision-making processes involving customers, partners, and firms. It needs a digital platform to monitor the value created in the customer ecosystem continually, so that the firm's capability becomes one continual exploration of distinct activities of value creation, instead of making a radical change in business models when there is a significant performance decline. Hence, the first research question:

R1: What capabilities are needed for the business value cycle to adapt to a changing customer value cycle?

Such capabilities, however, vary depending on the length of the business value cycle, as businesses producing physical products with a longer development cycle cannot develop new products as fast as those that develop service products. Similarly, the degree of influence the customer ecosystem (e.g., culture, socio-economic status, and other social determinants) has on the customer value cycle calls on firms to create new value propositions quickly through service offerings. Hence, the follow-up research question is:

R2: Are the capabilities needed for business value cycle adaptation different for customers with longer or shorter value cycles?

These questions are investigated in this paper. The next section uses a value lens to provide a framework and explicate the role various firms can play along the value continuum to compete. It identifies distinct activities, such as engaging customers, enlisting partners, expanding services, and enabling value creation. Section Three looks at how three organizations from published research have built agility to engage customers using these activities and how they have had extended their ability to remain agile during COVID-19. Section Four highlights key insights: customer empowerment,

collaborative competency, and operant IT resources, as capabilities to support e-relationships using digital platforms. Section Five discusses how two firms have leveraged these capabilities to remain agile during COVID-19, based on in-depth interviews conducted in 2020. Section Six provides recommendations for building e-relationships through a customer lens. The last section provides some concluding comments and ideas for future research.

A FRAMEWORK TO LINK CUSTOMER AND FIRM VALUE CYCLES

The early part of digital transformation in the 21st century was mostly focused on leveraging e-commerce platforms to extend the firm's value chain activities to guide the customer's need for value fulfillment as they went through their purchase journey. The focus of the firm's digital transformation was on re-envisioning the role of customer interfacing activities, such as marketing, sales, and distribution activities pre- and during purchase, as well as services used to support maintenance post-purchase. However, customers are increasingly seeking to take control of more things using their resources, such as making their stock trades, doing their taxes, cooking, and cleaning while renting a home when they travel, and even having recipes and ingredients sent home to cook different meals. In later stages of digital transformation, this led firms to re-envision what value-fulfilling activities they can distribute to customers, leveraging technology and external partners in support of this effort. These changes have begun to deepen firms' engagement with its customers using digital platforms, potentially leading to co-creating value propositions that align both the customer's and the firm's goals. In many ways, this is like co-designing products with suppliers and partners, except that the relationship between firms and their suppliers/partners is managed through service level agreements. In contrast, the relationship between firms and customers is based on the quality-of-service expectations.

For further discussion on the type of digital transformations needed to create and sustain value through relationship building, let us use Figure 1. The customer value cycle is shown along the y-axis, and the firm's value cycle is shown along the x-axis. The length of both the customer's and the firm's value cycle are categorized as either short or long. We will look at each quadrant to discuss what firms that operate in that quadrant must do to create and sustain value.

Q1: Shorter Value Creation Cycle and Shorter Value Fulfillment Cycle

Firms in this quadrant have a shorter customer value cycle and a shorter business value cycle. Customers engage in repeat purchases of these products (e.g., travel, restaurant, and recreational services), and firms need to support such engagement. Firms can react with agility to develop new value propositions, fulfill them, and evaluate their perceived value in use to begin the next value cycle. Firms have several opportunities to interact with their customers during purchase and post-purchase. Customer feedback during both phases is especially important, as the decision of customers to repeat their value cycle with the firm is influenced not only by their personal experience but also by peers and alternative service offerings from competitors.

Firms often use coupons to support a future purchase if customers have an unhappy experience or use loyalty card memberships to increase value over repeat purchases. They also may elicit partners to develop complementary services to differentiate their value propositions in the short-run and expand their service offerings to broaden the customer experience. For example, a restaurant may provide reservations, home delivery, recipes for foods customers like and providing cooking lessons. In some cases, firms may enable customers to help co-create value with them. For example, firms can co-design toys or product packaging to address safety and portability. Health systems can co-design remote monitoring services with patients who prefer to self-manage their health by leveraging access to local community care providers (Goldkuhl & Perjons, 2014). Such value co-creation and use of partners in expanding service options start with the continued and sustained engagement with customers during their value fulfillment. The role of digital platforms that track and analyze customer behavior is widely discussed in the literature (Grover et al., 2018).

Q2: Shorter Value Creation Cycle and Longer Value fulfillment Cycle

Firms that produce commodity products (toys, books, commodity software, materials or parts, tools, fabric, oils, etc.) tend to have a shorter customer value cycle but a longer business value cycle to fulfill new value propositions. Also, they usually do not interact directly with the eventual consumers of their products and must co-create value propositions with intermediaries (large retailers, grocery industry, or home construction industry) to create differentiation. Advanced technologies are dis-intermediating many of the traditional intermediaries and replacing them with digital firms or product manufacturers. For example, books and software are distributed digitally, and detergents and replacement parts are distributed by service providers of products who remotely monitor value-in-use. This means that there is a constant need to elicit partners to co-create value for differentiation and get the products when needed to eventual consumers.

Most of these firms co-create new value propositions with intermediaries by collaborating with them to understand customer criteria, such as faster replenishment and disposal services (e.g., oils, paints, detergents, etc.), bulk purchase savings (e.g., based on the volume of purchases), or attractive packaging (e.g., easy storage or less spoilage). Advanced digitization of eventual consumer and intermediary value cycles means that there is a greater opportunity to engage with consumers downstream to better understand customer needs. For example, customers using wearable devices to monitor their health conditions or engaging with peers using digital exchanges after cosmetic surgery (Khuntia et al., 2017). Such digital exchanges can help not only clinicians (intermediaries) but also the device manufacturers or digital exchange operations, so that they can expand service offerings to address patient needs. Timely tracking and adaptation of service offerings are key for differentiation in a commoditized marketplace (Ives et al., 2016).

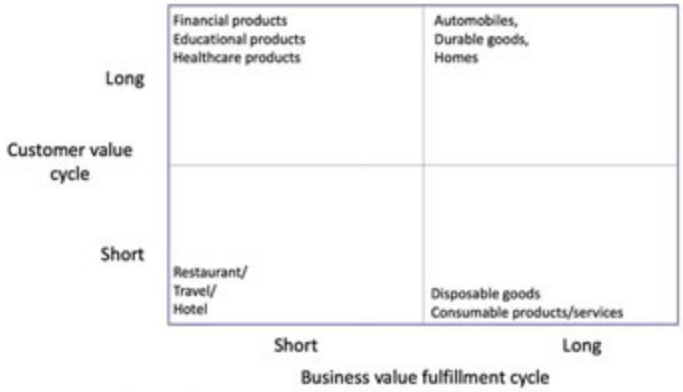

Figure 1. Value Lens: Customer and Business Value Cycles.

Q3: Longer Value Creation Cycle and Shorter Value Fulfillment Cycle

The customer value cycle for these firms tends to be longer given the inherent nature of the service contracts (e.g., insurance and financial products) and/or significant switching costs customers incur if they change the services during their contract period. However, by virtue of them being knowledge-intensive, these firms tend to have a shorter business value cycle. Due to the influence of evolving technology and regulatory changes within this business value cycle, it takes a longer time to develop new value propositions but less time to fulfill them. For example, tax and investment firms must spend time to understand changing laws and regulations to create value propositions, but they have a shorter period to decide which parts of customer software and company software must change to address these changes.

Given the longer-term involvement with customers during value-in-use, firms here can engage with customers during their value-in-use, enable customers to collaborate with the firm in co-creating value based on their distinct needs, and create new services (Cronholm et al., 2020). For example, a utility company that provides power may recognize the need for differentiation in services in certain neighborhoods for flexible or deferred pricing or to address their focused needs on conservation. This can lead to having them elicit partnership work with non-profit organizations to manage underserved communities or work with firms that weatherize homes for efficiency. Today, health systems are continually interacting with patients to help co-create distinct services that will enable customers to self-manage their health and use local resources to gain access to care, while soliciting partners to help secure these resources (Tanniru et al., 2018). The need for IT platforms, especially distributed platforms, to connect diverse actors and share data is critical to creating transparency and coordinating activities in such environments (Peral et al., 2020).

Q4: Longer Value Creation Cycle and Longer Value Fulfillment Cycle

A longer value cycle is typical of firms that produce durable products like automobiles or appliances, and these products also have a longer customer value cycle. While the firms cannot design new products quickly, they can use engagement with customers during their product use to look for opportunities to identify new services that will complement their products. Some of these include remote maintenance, automatic replenishment of parts, adding infotainment and security services in cars. Of course, many of these services call on these product companies to elicit partnerships with others, such as technology providers like Apple or Google for infotainment services, Zipcar for rental services, and sensor tracking services for remote home monitoring.

Sustained engagement with customers will help enable customers to co-create value propositions (e.g., sustainable disposal of unused detergents or refilling prescriptions based on prescription bottle use for seniors). It will also help expand service offerings by having partners provide some of these services as part of home maintenance or patient adherence to physician treatment plans. An IT platform that supports continual customer interaction, specifically during value-in-use, is key to discovering opportunities for new value propositions (Bilgeri et al., 2019).

In summary, the key activities that firms need to focus on within each quadrant, as they engage in digital transformation, are shown Figure 2. While all firms engage in these activities to support their value cycle and to broaden their reach into the customer value cycle using IT platforms, the initiating activity is what is shown in each quadrant to build customer relationships. Engagement during customer value fulfillment is the driver to create value for service firms with short value cycles in Q1, and enabling customers to use their experience during value-in-use to co-create value propositions is key for firms in Q3. Similarly, eliciting partners is often the beginning for firms in Q2 to learn about consumer needs to develop differentiating services, and expansion of service offerings to complement the product is key for product firms in Q4. No matter where a firm is situated, learning about the customer journey and adapting the business value cycle quickly as customer expectations change, is critical. It requires agility in performing each of these activities--engaging and enabling customers in defining value propositions that align with customer and firm goals, expanding services, and eliciting partnerships to help the firm fulfill these value propositions. We will explore how three firms have used these activities to build agility within their organization prior to and during COVID-19 in the next section.

BUILDING AGILITY USING VALUE CYCLE ACTIVITIES--REFLECTIONS FROM THREE CASE STUDIES

Extending the business value chain into customer decision-making is a competitive necessity for firms today, but it comes with dual challenges. While it provides a window into the customer ecosystem, firms need to use this window to engage and enable customers to co-create value propositions that align their shared goals and help the firms use agility to expand services and elicit partners to fulfill

these value propositions. We will briefly discuss three cases from different industries to see how they have supported such agility to stay competitive.

Case Study 1: LEGO (Sawy et al., 2016)

LEGO provides both young and old with interlocking blocks to build unique artifacts, and the firm has significant brand recognition. On the surface, it can be viewed as a commodity product company in Q2, with a longer value cycle of its own while addressing the needs of consumers with a shorter value cycle. It takes a longer time to develop new blocks and artifacts that use these blocks, and it produces them in bulk for economies of scale. However, consumers frequently purchase these products to address specific needs--provide a gift, expand their creations, etc.--and use retail intermediaries to purchase them.

Given LEGO's need to distribute its product through retailers, the company enlisted its largest retailers (their direct customers) to participate in product development, help forecast customer trends in purchasing games, and develop product replenishment strategies to align with consumer behavior. This led LEGO to give Stock Keeping Units (SKUs) to large retailers on an exclusive basis, and it used its technology platforms to manage retailer inventory and co-develop new products by empowering its employees. Empowered employees were able to extend their reach to LEGO's consumers beginning in 2008 by enabling LEGO fans to design new models (i.e., co-creation of value) using a separate digital platform (https://ideas.lego.com/). Fans voted on new design ideas, and the top ideas were sent to the LEGO review board for evaluation. Selected designs were developed as products, with project creators receiving 1% of net sales. The digital platform supported not only the co-design of products with customers but also the co-development of inventory and placement strategies with retailers, especially by developing unique paths to micro-market products for loyal customers.

In summary, instead of viewing itself as a commodity firm in Q2, LEGO enabled consumers to co-create value and collaborated with both retailers and its fans in value fulfillment by designing and distributing products to where and when customers needed them. By building brand loyalty and sustaining customer engagement through repeated customer value cycles, LEGO is operating as if it is in Q3.

Case Study 2: Marriott (Nyangwe & Buhalis, 2018)

Marriott, like many other hotel chains, is a commodity service company that provides short term stays for traveling consumers (i.e., operating in Q1). With many competitors and new non-hotel type firms (e.g., Airbnb providing temporary residency services for some consumers), Marriott needed to continue to engage customers. It began using its flagship rewards program and empowered Nick Johnson (the Pokemon Go enthusiast) to catch the remaining three Pokemons and complete his quest to "catch 'em all." Using customer interest and passion, Marriott began to have social conversations to form a bond with a select group of its loyal customers (called Marriott Insiders). Further, it enabled them to co-create new value propositions using their experience with their hotel stay. Some of these allowed customers to explore new destinations during their stay and share their experiences with others using social media.

Marriott empowered its employees to collaborate with multiple stakeholders to fulfill new propositions, such as using geo-fencing to present content as M Live experiences happening on Marriott property or delivering special desserts to guests for their birthday on-property. It elicited partnerships with technology providers to create platforms that supported both a mobile marketing strategy for geofencing and location-based engagement of targeted social media users (e.g., HYP3R).

In summary, by enabling select customers to help co-create new value propositions using a digital platform and eliciting partners to expand its service offerings. Marriott appeared to function as if it is in Q3 by building loyalty and increasing the frequency of its customer visits when they travel.

Case Study 3: Kaiser Permanente (Kagan et al., 2016)

Kaiser Permanente (KP), a healthcare provider, can be viewed as a firm with a longer customer value cycle and a shorter business value cycle as it fulfills the care-related needs of its patients while they are in the hospital. In other words, KP is positioned in Q3. Given regulatory and payment policies as well as the affiliation of physicians with hospitals, patients stay with a healthcare provider such as KP for at least one year or more. This provides KP with opportunities to engage patients, physicians, and other stakeholders using a digital platform, HealthConnect, to create new value propositions. Sharing healthcare data securely via kp.org improved care continuity by having its physicians' groups engage patients at home. For example, the digital platform allowed 70% of KP customers to improve their adherence to care-related treatment plans.

KP empowered its physicians to use digital tools to support their clinical work, such as remotely monitoring patients' blood sugar levels to manage diabetes. It also leveraged technologies to assess patients based on risk criteria, such as diagnoses, visits, and adherence to pharmaceutical regimes, for tailored care. Moreover, it used social media-enabled care circles to provide patients with peer support; all supported using partnerships with advanced technology vendors.

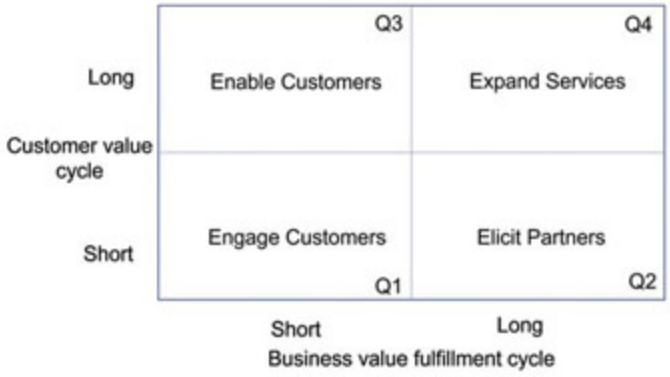

Figure 2. Key Activities to Build and Sustain Value Cycles

In summary, by virtue of it being both a provider and payer, KP can use multiple value propositions to deliver care to patients and keep both its physician and patient population loyal, thus remaining in Q3.

A review of these three cases is summarized in Figure 3. Even though each of these firms began in different quadrants, they all used various activities to align their goals with those of their customers, thus acting as if they are in Q3. Digital platforms play a key role in supporting the agility with which firms can use their extended reach into the customer value cycle to create and sustain the value of multiple value cycles. This became even more critical during the recent health emergency (COVID-19), as firms needed to co-create value propositions with many other actors in the customer value cycle and expand their services by soliciting from some of these actors. Some of these are discussed briefly in Table 1 (Brown, 2020; Grants strengthen COVID-19 response in homeless communities, 2020; Marriott Cleanliness Council, 2022; Huang, 2020; Ruvin & Khurana, 2020).

The need for agility to address such health emergencies was apparent not only for firms in Q1, Q2, and Q3, but it also became critical for those in Q4. In fact, many product manufacturing firms have had to expand service offerings for customers who could not visit showrooms or home showings in addition to employees who could not work in their offices. When employees had to come to work, these firms needed to elicit partnerships with new vendors to build barriers to support social distancing and provide testing and even vaccination services for those who visit their work areas. In other words, besides traditional mechanisms to engage and enable customers to help develop value propositions

for services that complement products, and to elicit partners to fulfill these services, firms needed agility to address evolving market dynamics by extending their reach into customers and partner ecosystems. This leads us to the discussion on key insights in organizational capabilities that support organizational agility beyond internal governance, structure, and processes. This is discussed in the next section.

KEY INSIGHTS--ORGANIZATIONAL CAPABILITIES

In a networked economy, actors from multiple ecosystems (customer, firm, and partners) are asked to bring a mix of their knowledge and skills, along with their resources, to address changing customer demand for products and services (Vargo & Lusch, 2004; 2008). Each of the actors operates under different rules, norms, and a culture associated with their own ecosystem. Digital platforms can help extend the firm's reach into the ecosystems outside their organizational boundaries to co-create value by leveraging the niche competencies of customers and partners (Prahalad & Ramaswamy, 2004; Xiao et al., 2011). However, digital platforms cannot by themselves support the organizational agility needed to align the goals and actions of all involved. From the discussion in the previous section, we draw three distinct capabilities in this extended ecosystem. These are shown in Figure 3 below and discussed further in this section.

Customer Empowerment

Empowered customers become active in co-creating value that reflects their needs as they try to fulfill them within their ecosystem using the actors and resources that they have access to (Moller, 2006; Moller et al., 2005). Such empowerment gives customers a sense of control over their contribution to the firm's product/ service offerings (Bolton & Saxena-Iyer, 2009; Fuchs et al., 2010). Firms can use different strategies, such as crowdsourcing, user innovation, brand communities, and mass customization practices, to support this effort (Acar & Puntoni, 2016; Priharsari & Abedin, 2021). In healthcare, such empowerment can help customers gain control over their health management by acquiring the needed skills to achieve health outcomes (Khuntia et al., 2017) and have a positive attitude toward the services provided by the firm (Sembada, 2018). When customers must combine resources from multiple actors within their ecosystem to complement the products provided by the firm, such empowerment can help integrate these resources (Alt et al., 2019), especially if the firms' digital platforms can support such integration (Fuchs et al., 2010).

As discussed in the three cases and elaborated in Table 1, each company used its digital platforms to help with product design ideas. Companies get enthusiastic customers to engage in games that excite other customers and enable them to share their experiences with the community, empowering both internal actors and patients to work towards changing health behaviors. During COVID-19, they leveraged other actors in the customer ecosystem to co-create value to continue to sustain value to their customers.

Collaborate Competencies

Firms today need to collaborate with several partners, including customers. Building competencies in developing new partnerships with agility is key for creating new value propositions quickly. Besides having skills in creativity, imagination, and innovation (Beresford & Trevillion, 1995), firms need agility to collaborate with other firms, so that partnering firms can jointly invest in resources to build the capacity needed (Lee et al., 2020). Such collaboration may lead to co-location to improve agility in manufacturing (Sawy et al., 2016), as well as cross-functional and inter-disciplinary skill development through knowledge sharing (Wulf et al., 2017). Collaborative competencies support the notion that different teams can work together and complement their skills in competitively positioning the firms for collaborative advantage (Mrass et al., 2021).

This is evidenced in the following: (a) young customers were inspired to innovate and compete, as well as partner with the firm's actors in product development (LEGO); (b) used a common digital

platform to share experiences so they could collectively lead to an improved experience for all in the future (Marriott); and (c) turned technology providers into partners in an evolving customer ecosystem influenced by changing patient capacity and interest in managing their own health (KP). Again, the firms must use their collaborative competencies to work with other newer partners to support the design of services needed to expand the service offerings.

Intelligent Digital Platform (IT as an Operant Resource)

Much has been written about the role of digital platforms in integrating the resources and actors of both the firm and customer ecosystems and the need for their agility, as the competencies and capabilities of actors evolve over time. The platforms need to put these competencies together collaboratively to build the agility that firms require to address the needs of customers. With the increased liquefication of resources to support the virtual design and development of value among multiple actors, IT is no longer just a communication platform. It is an intelligent platform with autonomous agents using analytics on stored knowledge to increase the agility with which a firm can respond (Barrett et al., 2015). These agents can make the platforms dynamic by complementing the skills of human actors and complementing organizational resources by synthesizing both structured and tacit knowledge using some new AI tools (Li & Chan, 2019). Such a capability can not only help supply chain and partner collaboration in developing new products (Ayala et al., 2020), but it can also support demand side collaboration with customers to respond to certain needs with agility. For example, Internet of Things (IoT), natural language processing, and AI tools have enabled intelligent agents to support nursing staff in addressing the needs of delirium patients with agility (Tanniru et al., 2021). In other words, IT is an operant resource to create and fulfill value just like human actors. Intelligent platforms allow the participation of a broader group of stakeholders with heterogeneous knowledge and capabilities to create and sustain value (Gawer & Cusumano, 2002). Such platforms can inspire, mobilize, and maximize the distributed skill and knowledge of all actors with agility using a layered modular architecture (Yoo et al., 2010).

LEGO used its digital platform to accumulate service interactions with customers to develop new ideas. Marriott used its platform's stored knowledge to align the goals of staff and customers in creating new value propositions, and KP used its knowledge platform to identify high-risk patients to tailor care on-demand. Again, each has leveraged their digital platforms and their knowledge in co-creating value or design of services to quickly meet the challenges during COVID-19.

In the next section, we will use the insights on these capabilities and discuss how two other organizations leverage them to remain agile during COVID-19.

CAPABILITIES TO SUSTAIN E-RELATIONSHIP DURING COVID-19

In this section, we summarize how two organizations used the capabilities they developed prior to COVID-19 to adapt to disruptions that have occurred during COVID-19.

TSMC--A Manufacturing Company--Operating Between Q2 and Q4

Taiwan semiconductor manufacturing company (TSMC) provides advanced logic technology semiconductors as base products used by its customers: fabless semiconductor companies. These business customers create distinct value for their customers by providing the high-power processing efficiency, increased speed, and complex functionality needed in today's digital marketplace. The customers tend to repeat their purchases several times, anticipating new value with each new purchase. Such value is needed to complement innovations created by downstream support chain actors in the creation of several end products, such as mobile devices, high-performance computing processors, automotive electronics, and Internet of Things (IoT).

Table 1. Summary of activities of three organizations before and during COVID-19.

Key Insights	Engagement and Enablement of Value Co-creation		Extended services and Elicitation of Partners	
	Activities Pre-COVID-19	Activities during COVID-19	Activities Pre-COVID-19	Activities during COVID-19
LEGO	Design a special website for a fan group Provide meaningfulness to the product development Generate competence in fan belief of value creation	Develop the activity #letsbuildtogether to share building ideas Develop a new Sesame Street special episode to support and address children's feelings about COVID-19	Collaborate with young customers and create competition among teams Build selected products for commercial-ization. Reward project creator	Extend partnership with Tencent to offer more creative and safe digital play experiences for children Partner with UNICEF to support parents and communities to provide care and playful learning opportunities Partner with Pinterest to create activity-related content to support those stuck at home during COVID-19
Marriott	Align its reward program to empower enthusiastic players Participate and share special events of customers	Implement touchless services where guests can check in, access their rooms, make special requests, and order room services without contact Allow guests to cancel reservations up to 24 hours prior to travel	Partner with geo-fencing technology provider Collaborate with customers to enrich their overall experience on its property	Develop distinct partners to provide services that allow touchless disinfecting capabilities Offer food delivery services in many international locations

(*Continued*)

One can view TSMC as a commodity firm, positioned somewhere between Q2 and Q4, with a long value fulfillment cycle supporting other customers and with a long value cycle (operating between Q2 and Q4). The ultimate end customers of TSMC are those who purchase mobile phones, computers, etc. (with a short or long customer value cycle).

Empowering stakeholders: TSMC enables its customers to co-design products by understanding their technological needs. Typically, it takes more than a year for TSMC to develop new product offerings, and technology is used to co-design these products with its business customers. Co-creation of value is key given the huge investment of resources involved in product design and development. TSMC uses its employees (product innovators) to accelerate technology differentiation in the development of new product offerings.

Collaborative competency: TSMC uses an intelligent manufacturing environment to fulfill the value created, i.e., manufacture the products designed. This environment leverages advanced technologies such as artificial intelligence, machine learning, expert systems, and advanced algorithms, to capture knowledge and support knowledge sharing among various actors (machine and human) in manufacturing the products. The new products not only use an intelligent manufacturing environment to manufacture the product, but they also leverage empowered TSMC employees to address changes in customer needs quickly, such as by responding to urgent pull-in requests.

This intelligent manufacturing environment evolved from manual operations in 1987, to semi-autonomous mode in 1993, to fully automated mode in 2000, and finally to intelligent mode using machine learning in 2014-2015. Currently, this environment uses deep learning that evolved from rule-based algorithms of the past, where heuristics from domain experts are explicitly coded by software developers to make decisions. Today, machine learning models use massive, real-time data to generate outputs.

Frank Lin, director of the business system integration division, says, "Machine learning models have increased the accuracy from 90% (rule-based systems) to 97%."

Intelligent digital platform: To support the co-creation of value, TSMC uses the Open Innovation Platform[R] (OIP) cloud alliance, as it provides its customers with a safe and secure environment for collaborative design. In addition, it uses IT as a valuable resource to co-design products with partners (eFoundry[R]) and creates differentiated value for customers. For example, TSMC-Direct[SM] connects partners for mission-critical business processes and uses information sharing to give supply chain visibility, reduce cycle time, and manage inventory levels and costs. It uses the TSMC-Online[TM] communication platform to interact with customers all through the product value cycle by sharing information on product design, tape, and wafer tracking. The goal of these digital services is to improve collaborative competencies in design (reduce time-to-market), engineering (reduce time-to-volume), and logistics (reduce time-to-delivery).

"The functionality of such a 24x7 real-time information sharing platform helps customers to provide feedback at a faster pace and shorten the value fulfillment cycle of new product offerings." Greg Chang, department manager of the sales system integration department.

Agility is key to building relationships with customers and developing a competitive strategy for TSMC. Even though TSMC has the capability to design and manufacture the products its customers make, it chose to provide components for customers so they can manufacture the semiconductors. Such a philosophy of "not competing with customers" has made TSCM a favored foundry partner--a key customer relationship strategy. While COVID-19 has not directly impacted TSMC's interaction with its customers, its capabilities continued to support changes in customer expectations.

In summary, TSMC has developed an organizational capability to empower its customers so that they are continually engaged in support of customer value creation, and the firm used its collaborative competency to elicit partners and expand service offerings as needed--all supported by a dynamic digital platform.

Authority Health--A Primary Care Facility-Operating Between Q1 and Q3

Founded in 2004, Authority Health (AH) is a Public Body Corporate organization that collaborates with community health centers and governmental and nongovernmental organizations to strengthen the health care safety net of Detroit, Wayne County, and the Southeast Michigan region. AH manages the nation's second-largest primary care graduate medical education teaching health center program. It also provides several services for its patients: health insurance enrollment and navigation services, community-based primary care services, and a health data portal to share information among all its clients. AH owns and operates the Popoff Family Health Center, which provides primary care to residents of Detroit's east side while training family medicine residents in this community setting. Popoff accepts all patients, all health plans, and the uninsured. AH also manages Pediatrics, PLC, a site providing a full range of services for children while training residents in pediatrics. In a way, AH has two clients: medical residents who are getting trained to service patients in underserved communities and patients who need tailored care so they can overcome health inequities.

Empowering stakeholders: To reach vulnerable populations in delivering care, AH has developed relationships with institutions in the community so that they can be empowered to help support vulnerable populations. These stakeholders include Black Family Development (supports families who

intend to stay together); Brilliant Detroit (dedicated to building the success of kids, families, and neighborhoods); Detroit Public Schools; Fresh Rx (provides fresh fruit and vegetables to needy members of the community); and Project Health Community (addresses nutritional needs of school-age children). Based on stakeholder input, AH has begun to develop services for the homeless and support programs to address childhood development.

Collaborative competency: Using collaboration with its clinical partners, Michigan State University College of Osteopathic Medicine (MSUCOM), and other community agencies, AH supplies physicians to Detroit and the region, including federally qualified health centers (FQHC) that provide care to vulnerable populations. These centers include Detroit Community Health Connection, Western Wayne Family Health Center, Family Medical Centers of Michigan, and Covenant Community Care. Resident physicians complete a certificate in Population Health (in collaboration with the University of Michigan), which is integrated with a Community Medicine block rotation and clinical quality improvement projects designed to address social determinants of health and disparities in health and health care. Such resident engagement in these communities is designed to help them choose to establish a practice upon graduation. As of today, two-thirds of the AH graduates establish their practices in primary care and health professional shortage areas.

Agility to address COVID-19 has impacted populations with significant health disparities (racial/ethnic minorities, those with less privileged socioeconomic status) and medically and/or socially vulnerable populations (residents of chronic care facilities, community-dwelling older adults, pregnant women, children, persons with cognitive impairment or dementia, homeless, incarcerated). Similarly, it has impacted those care providers who service these populations, including medical personnel who direct patient care and staff that service patients. For example: office personnel, home health aides, family and informal caregivers, emergency responders, and frontline workers in essential businesses or services.

"We at AH continue our commitment to underserved and vulnerable populations of Detroit during these challenging times. We used virtual technologies to collaborate with our community partners such as Project Healthy Community and Brilliant Detroit with information sharing and engaging in group and individual consultations." Dr. Ernie Yoder, Director of Medical Education at AH.

Using its collaborative competency with the community, FQHCs, and local health systems, AH can address many challenges faced by both its clients (patient population and care delivery personnel). In addition, AH leveraged its core competency (medical education) to create new processes and prepare materials requisite to address the challenges presented by the COVID-19 pandemic. It developed screening procedures for safe in-person clinical encounters, drive-up testing for COVID-19, and adapted standard infection protocols to protect all personnel at the various clinical sites. New signage was developed for clinical sites to assist patients in navigating the redesigned clinical environment. It must empower its many stakeholders to learn about safety protocols and gain access to personal protective equipment (masks, shields, gowns, gloves, etc.) to address both their safety as well as those for whom they provide care. They are also asked to identify/access testing facilities to ensure all the staff in its Popoff Family Health Center are tested for COVID-19 and support the local community who lack access to such testing.

"Following technology assessments and a rapid planning phase, our lead physicians at our three practice sites facilitated the changeover from face-to-face patient encounters to remote, utilizing Doxy.me." Dr. Yoder.

Intelligent digital platform: By adding telemedicine services to connect with patients who ordinarily would receive care through physical visits to the physician's office and leveraging medical knowledge on COVID-19 to revise portal information in support of education, share test results, and answers questions, AH was able to make its digital platform dynamic and adapt to the new reality. While printed material will continue to be developed for those who do not have access to technology as well as to

support wider dissemination to people at churches, workplaces and the like, AH is able to provide valuable service to the community at large.

In summary, AH has begun to co-create value with new services to support its clients and leverage its collaborative competencies to bring on current as well as new partners in fulfilling some of these services. It has begun to enhance its digital platform to support virtual learning and consultation. Many of these new services will continue to sustain the value to the vulnerable populations of the Detroit region as well as to medical students and residents. Table 2 discusses some of the individual services used to create value pre-COVID-19 and how these are transformed to address the challenges during COVID-19 to sustain the value.

RECOMMENDATIONS FOR BUILDING E-RELATIONSHIPS THROUGH CUSTOMER LENS

Table 1. Continued.

Kaiser Permanente	Develop a digital platform to allow access to healthcare data Improve self-determination to change work behaviors	Implement "virtual first" strategy for non-COVID-19 patient care Use digital platform to access COVID-19 information proactively Waive costs for member patients receiving COVID-19 treatment	Partner with technology vendors to build portals used by customers for self-management of their health	Partner with skilled nursing facilities, residential care facilities, and families at home to provide care with remote monitoring Gather data from EHR and biometric information in each locale and share it with public health officials Partner with National Health Care for the Homeless Council to address the needs of homeless populations Partner with CDC and ten other organizations to develop evidence based best practices to strengthen the public health infrastructure to counter COVID-19 Partner with Dignity Health, State of California, and Los Angeles County to create additional beds and ICU capacity

Three published case studies provided key insight into how firms in the digital age need to extend their electronic commerce platforms to build relationships with customers by leveraging advanced technologies in customer value creation and sustaining this value creation using repeated value cycles (creation, fulfillment, and assessment during value-in-use) as customer expectations evolve over time. To support such adaptation to create and sustain value, firms need to play four digital roles: engage customers, elicit partners, enable value co-creation, and expand services. Organizations from three different industries (LEGO, Marriott, and Kaiser Permanente) were used to show how they used these

digital roles to build capabilities to create and sustain value. These three organizations and two others, analyzed in-depth, were used to illustrate these digital roles even during a health emergency (COVID-19). The two new organizations are TSMC, a supplier of chips to digital device manufacturers in Taiwan, and Authority Health, a primary care provider in Detroit, Michigan. This analysis leads to our following recommendation: independent of customer value cycle frequency (frequent or less frequent) and the length of an organization's value fulfillment cycle (short or long), these digital roles can help transform an organization to remain agile in an evolving market dynamic. Such a transformation, however, calls on organizations to build three key capabilities discussed below.

Empower Stakeholders (Customers and Employees)

Empowered customers are influenced by advances in technologies and sensitive to innovative niche services brought into the market by digital competitors. Organizations operating in a knowledge-intensive and service-driven economy must not only empower customers through the co-creation of value but also empower employees and external partners to innovate and use a mix of internal and external resources quickly to stay competitive (Moller, 2006; Moller et al., 2005; Sunyaev, 2014). LEGO empowered its customer community by having them compete on new product design ideas using specially developed web platforms. Marriott empowered Nick Johnson, the Pokemon Go enthusiast, to catch the remaining three Pokemons to complete his quest to "catch 'em all," thus creating customer enthusiasm. KP developed a digital platform, HealthConnect, to empower its clinical staff and partners to access relevant patient healthcare data so patients are empowered to self-manage their care. TSMC empowered its customers and employees to co-design high-tech products that offer differentiating features. Authority Health empowered its staff and local communities to adapt their workflows using technologies to support continued care of its vulnerable patient populations during COVID-19. Each of these organizations recognized the role customers, employees, and partners can collectively play in helping organizations adapt their value cycle activities quickly to changes in the competitive landscape and customer expectations. Such a collective engagement of actors from different ecosystems cannot occur unless organizations build the next capability: collaborative competency.

Build Collaboration Competency

While organizations can empower customers in value creation, employees must learn to adapt their internal operations quickly and leverage external partners for needed resources to design and develop new services to fulfill the value. LEGO recognized its young customers' creativity and allowed them to collaborate with internal development staff to transfer customer ingenuity and competitive spirit into competitive products. Marriott collaborated with a geo-fencing technology partner to capture customer experiences and used the HYP3R platform to help collaborate among staff and customers on its premise to enrich customer experience. KP collaborated with technology vendors to build portals that allowed collaboration between patients and staff to track health conditions and give patients the ability to self-manage their health. TSMC developed an intelligent platform to allow its employees to collaborate with customers in making design changes quickly and shortening product delivery time. Authority Health used the collaboration of its staff, community partners, and local health providers to react quickly to track gaps in patient care during COVID-19 and address them to reduce complications. In all these cases, organizations made collaboration a key competency so that multiple actors could design and develop new service innovations quickly to create value for their customers. Such collaboration to support agility is not possible without a dynamic digital platform.

Design an Intelligent Digital Platform with IT Becoming an Actor

Organizations have used digital platforms to integrate their resources with partners to innovate new products/services. While not all such innovations are commercially viable, they will support organizational learning if knowledge from past failures and the context of these failures become part of the organizational knowledge base. Organizations engage customers in value cycle activities, and not all customers complete these activities (i.e., they may interrupt their customer journey) for any number

of reasons. Advanced technologies are helping organizations gain insight into customer behaviors by analyzing their digital interactions with the firm. Digital advances are increasing the liquefication of resources (i.e., digital data captured in multiple interactions) and resource density (i.e., the number of external actors participating in these interactions) (Barrett et al., 2015) to facilitate faster development of innovative products and services. With big data analytics and machine learning, IT is becoming an operant resource (knowledge actor) by allowing organizations to leverage prior knowledge about failures and successes. Further, IT provides the knowledge of customer behaviors and enables virtual product design to speed up the design and development of new value propositions, thus making the digital platform that connects all the actors learning oriented or dynamic. This was seen in some of the cases discussed in this paper such as modular design of products (e.g., LEGO blocks or components of TSMC), analysis of customer behavior (e.g., patient experiences in KP and customer experiences in Marriott) and adapting to changing market dynamics (COVID-19 impact on patients with co-morbidities by AH and niche product designs by TSMC). In other words, organizations utilize digital platforms to play the role of an actor (an operant resource) in designing and developing new value propositions.

CONCLUSION

Organizations have sought to transform their operations by building agility to react to a changing customer and technology landscape. It has been suggested that they may need a separate organizational model that is agile, with its own governance, structure, and processes, to explore and evaluate new innovative value propositions. With changes occurring at a faster pace and customers seeking differentiated products/services, organizations are continually challenged to segment markets for differentiation and even create new business models to stay competitive. Reacting to these changes should not be delegated to a particular group that coordinates the agile model, but agility to change should be part of the organizational strategy by building resiliency and capacity building. Such capacity building begins with expanding organizational actors (i.e., adding customers, partners, and technology actors to employees). Empowering actors to integrate their resources through collaborative competencies and use a dynamic digital platform that supports learning from each value cycle to create and sustain value and support customer relationship building.

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REFERENCES

Acar, O. A., & Puntoni, S. (2016). Customer empowerment in the digital age. Journal of Advertising Research, 56(1), 4-8. https://doi-org.ezproxy.hcplc.org/10.2501/JAR-2016-007

Alt, R., Ehmke, J. F., Haux, R., Henke, T., Mattfeld, D. C., Oberweis, A., Paech, B., & Winter, A. (2019). Towards customer-induced service orchestration--requirements for the next step of customer orientation. Electronic Markets, 29(1), 79-91. https://doi-org.ezproxy.hcplc.org/10.1007 /s12525-019-00340-3

Ayala, N. F., le Dain, M. A., Merminod, V., Gzara, L., Enrique, D. V., & Frank, A. G. (2020). The contribution of IT-leveraging capability for collaborative product development with suppliers. Journal of Strategic Information Systems, 29(3), 101633. https://doi-org.ezproxy.hcplc.org/10.1016/j .jsis.2020.101633

Barrett, M., Davidson, E., Prabhu, J., & Vargo, S. L. (2015). Service innovation in the digital age. MIS Quarterly, 39(1), 135-154. https://doi-org.ezproxy.hcplc.org/10.2307/26628344

Beresford, P., & Trevillion, S. (1995). Developing skills for community care: a collaborative approach, Medicine, DOI:10.3109/13561829809014104

Bilgeri, D., Gebauer, H., Fleisch, E., & Wortmann, F. (2019). Driving process innovation with IoT field data. MIS Quarterly Executive, 18(3), 191-207. https://doi-

org.ezproxy.hcplc.org/10.17705/2msqe.00016

Bolton, R., & Saxena-Iyer, S. (2009). Interactive services: A framework, synthesis and research directions. Journal of Interactive Marketing, 23(1), 91-104. https://doi-org.ezproxy.hcplc.org/10.1016/j .intmar.2008.11.002

Bossert, O., Laartz, J., & Ramsoy, T. J. (2014). Running your company at two speeds. McKinsey Quarterly, 4, 12-14.

Brown, M. (2020). Innovative partnership creates Los Angeles surge hospital. Kaiser Permanente. https://about.kaiserpermanente.org/our-story/news/announcements/innovative-partnership -creates-los-angeles-surge-hospital

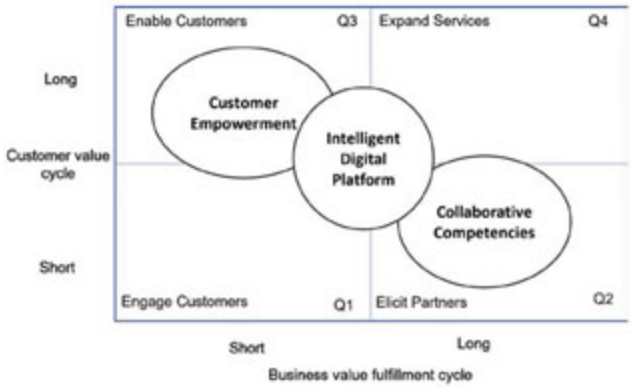

Figure 3. Key insights on organizational capabilities

Cronholm, S., Gobel, H., & Akesson, M. (2020). ITIL compliance with service-dominant logic. e-Service Journal, 11(2), 74-79. https://doi-org.ezproxy.hcplc.org/10.2979/eservicej.11.2.03

Fuchs, C., Prandelli, E., & Schreier, M. (2010). The psychological effects of empowerment strategies on consumers' product demand. Journal of Marketing, 74(1), 65-79. https://doi-org.ezproxy.hcplc.org/10.1509 /jmkg.74.1.65

Gawer, A., & Cusumano, M. A. (2002). Platform leadership: How Intel, Microsoft, and Cisco drive industry innovation (Vol. 5). Harvard Business School Press.

Goldkuhl, G., & Perjons, E. (2014). Focus, goal and roles in e-service design: Five ideal types of the design process. e-Service Journal, 9(2), 24-25. https://doi-org.ezproxy.hcplc.org/10.2979/eservicej.9.2.24

Grants strengthen COVID-19 response in homeless communities. (2020). Kaiser Permanente. https://about.kaiserpermanente.org/community-health/news/grants-strengthen-covid-19 -response-in-homeless-communities

Gronroos, C. (2008). Service logic revisited: who creates value? And who co-creates? European Business Review, 20(4), 298-314. https://doi-org.ezproxy.hcplc.org/10.1108/09555340810886585

Grover, V., Chiang, R. H. L., Liang, T. P., & Zhang, D. (2018). Creating strategic business value from big data analytics: A research framework. Journal of Management Information Systems, 35(2), 388-423. https://doi-org.ezproxy.hcplc.org/10.1080/07421222.2018.1451951

Huang, P. (2020). The LEGO group extends strategic partnership with Tencent--Providing more creative and safe digital play experiences for Chinese children. The Lego Group. https://www.lego.com /en-it/aboutus/news/2020/january/lego-group-and-tencent-partnership

Ives, B., Palese, B., & Rodriguez, J. A. (2016). Enhancing customer service through the internet of things and digital data streams. MIS Quarterly Executive, 15(4), 279-297.

Kagan, M. H., Sebastian, I. M., & Ross, J. W (2016). Kaiser Permanente: Executing a Consumer Digital Strategy, MIT Sloan CISR Working Paper (No. 408).

Khuntia, J., Yim, D., Tanniru, M., & Lim, S. (2017). Patient empowerment and engagement with a health infomediary Health Policy and Technology, 6(1), 40-50. https://doi-org.ezproxy.hcplc.org/10.1016/j .hlpt.2016.11.003

Lee, N. C. A., Wang, E. T. G., & Grover, V. (2020). IOS drivers of manufacturer-supplier flexibility and manufacturer agility. Journal of Strategic Information Systems, 29(1), 101594. https://doi .org/10.1016/j.jsis.2020.101594

Li, T. (Carol), & Chan, Y. E. (2019). Dynamic information technology capability: Concept definition and framework development. Journal of Strategic Information Systems, 28(4), 101575. https://doi-org.ezproxy.hcplc.org/10.1016/j.jsis.2019.101575

Marriott Cleanliness Council. (2022). Marriott. https://clean.marriott.com/

Moller, K. (2006). Role of competences in creating customer value: A value-creation logic approach. Industrial Marketing Management, 35(8), 913-924. https://doi-org.ezproxy.hcplc.org/10.1016/j.indmarman .2006.04.005

Moller, K., Rajala, A., & Svahn, S. (2005). Strategic business nets--Their type and management. Journal of Business Research, 58(9 SPEC. ISS.), 1274-1284. https://doi-org.ezproxy.hcplc.org/10.1016/j.jbusres .2003.05.002

Mrass, V., Peters, C., & Leimeister, J. M. (2021). How companies can benefit from interlinking external crowds and internal employees. MIS Quarterly Executive, 20(1), 17-38. https://doi-org.ezproxy.hcplc.org /10.17705/2msqe.00040

Nyangwe, S., & Buhalis, D. (2018). Branding transformation through social media and co-creation: Lessons from Marriott International. Information and Communication Technologies in Tourism 2018, 2018, 257-269. https://doi-org.ezproxy.hcplc.org/10.1007/978-3-319-72923-7_20

Peral, J., Gallego, E., Gil, D., Tanniru, M., & Khambekar, P. (2020). Using visualization to build transparency in a healthcare blockchain application. Sustainability (Switzerland), 12(17). https://doi-org.ezproxy.hcplc.org/10.3390/SU12176768

Table 2. Summary of activities of authority health before and during COVID-19.

	Pre-Covid	Under COVID-19	Partners	Post-Covid
Equipment for the staff	Not applicable	Work with physicians in health systems to acquire PPE, gowns, and handwashing, etc.	FQHCs and U.S. and international suppliers	Use this experience for planning resources in the future
Testing staff and patients	Not applicable	Set up testing centers and use local testing labs	Testing labs, and churches to host testing	Use testing experience to do preventive screening of patients for different disease conditions
Cleanliness and social separation	Standard cleaning and disinfecting protocols	Modify cleaning procedures, staggered work schedules, and set employee expectations		Use these protocols to reduce future infections
Patient access to staff	In-person access at primary care center	Use a hybrid model of telehealth and face to face visits	Tele-health software (Doxy.me)	Potential use to continue hybrid model to expand and facilitate patients access to primary care and consultation
Staffing resource flexibility	Not applicable	Use virtual interaction and telehealth to facilitate meetings and support those who are sick by having them stay at home	Health systems and medical students/ residents, technology providers	Providing flexibility to staff and use virtual services to improve productivity
Education to stakeholders	Procedures to interact with partners with roles and responsibilities	Create education material to address COVID-19 info and protocols for prevention	Public health agencies, CDC, and training specialist to prepare new material	Leverage the experience to reach patients and communities

(*Continued*)

Porter, M. E. (2010). What is value in health care? The New England Journal of Medicine, 363(26), 2477-2481.

Prahalad, C. K., & Ramaswamy, V. (2004). Co-creating unique value with customers. Strategy & Leadership, 32(3), 4-9. https://doi-org.ezproxy.hcplc.org/10.1108/10878570410699249

Priharsari, D., & Abedin, B. (2021). Orchestrating value co-creation in online communities as fluid organizations: Firm roles and value creation mechanisms. Information Technology and People. https://doi-org.ezproxy.hcplc.org/10.1108/ITP-10-2020-0707

Ruvin, H., & Khurana, A. (2020). Sesame workshop and the LEGO foundation to premiere Elmo's world news special worldwide to support families during the COVID-19 pandemic. Sesame Workshop. https://www.sesameworkshop.org/press-room/press-releases/sesame-workshop-and-lego -foundation-premiere-elmos-world-news-special

Sawy, O. A. E., Amsinck, H., Kraemmergaard, P., & Vinther, A. L. (2016). How LEGO built the foundations and enterprise capabilities for digital leadership. MIS Quarterly Executive, 15(2), 141-166. https://doi-org.ezproxy.hcplc.org/10.4324/9780429286797-8

Sembada, A. (2018). The two sides of empowering consumers to co-design innovations. Journal of Services Marketing, 32(1), 8-18. https://doi-org.ezproxy.hcplc.org/10.1108/JSM-02-2017-0062

Sunyaev, A. (2014). Consumer facing health care systems. e-Service Journal, 9(2), 1-23. https://doi.org/10.2979/eservicej.9.2.1

Tanniru, M., Khuntia, J., & Weiner, J. (2018). Hospital leadership in support of digital transformation. Pacific Asia Journal of the Association for Information Systems, 10(3), 1-24. https://doi.org/10.17705/1pais.10301

Tanniru, M. R., Agarwal, N., Sokan, A., & Hariri, S. (2021). An agile digital platform to support population health--A case study of a digital platform to support patients with delirium using IoT, NLP, and AI. International Journal of Environmental Research and Public Health, 18(11). https://doi-org.ezproxy.hcplc.org/10.3390/ijerph18115686

Vargo, S. L., & Lusch, R. F. (2004). Evolving to a new dominant logic for marketing. The Journal of Marketing, 68(1), 1-17. https://doi-org.ezproxy.hcplc.org/10.1300/J047v07n04_02

Vargo, S. L., & Lusch, R. F. (2008). Service-dominant logic: Continuing the evolution. Journal of the Academy of Marketing Science, 36(1), 1-10. https://doi-org.ezproxy.hcplc.org/10.1007/s11747-007-0069-6

Vargo, S. L., & Lusch, R. F. (2014). Service-dominant logic of marketing. In The Service-Dominant Logic of Marketing (pp. 61-74). Routledge.

Wulf, J., Mettler, T., & Brenner, W. (2017). Using a digital services capability model to assess readiness for the digital consumer. MIS Quarterly Executive, 16(3), 171-195.

Xiao, X., Sarker, S., & Rai, S. (2011). Product innovations in emerging economies: The moderating role Of collaborative capacity. https://www.researchgate.net/publication/221228768

Yoo, Y., Henfridsson, O., & Lyytinen, K. (2010). The new organizing logic of digital innovation: An agenda for information systems research. Information Systems Research, 21(4), 724-735. https://doi-org.ezproxy.hcplc.org/10.1287/isre.1100.0322

Zeithaml, V A. (1988). Consumer perceptions of price, quality, and value: A means-end model and synthesis of evidence. Journal of Marketing, 52(3), 2. https://doi-org.ezproxy.hcplc.org/10.2307/1251446

APPENDIX: RESEARCH METHOD AND MATERIAL

This research work is a mix of primary and secondary case studies. The first three case studies utilize the information available from already published cases and online material. The fourth and fifth applied primary case studies and extracted information by conducting interviews, annual reports, and company documents during 2020. The initial work of this research framework was presented at an international conference of service science and innovation in 2020 to get feedback from research scholars and practitioners.

The interview at TSMC was conducted by the senior professor, accompanied by two of his Ph.D. students and an associate professor from the same university. Responses for all interviews were handwritten. Before each interview, we acquired consent from the respondent to use their responses and quoted their real name and designation. The initial step was to present our value framework to senior management and ask them about how they have addressed disruptions imposed by COVID-19, and how they used capabilities, such as empowering stakeholders, collaborative competence, and dynamic digital platforms, to mitigate the impact of the pandemic. Table 1 contains the details of the method used for data collection.

Moreover, the interview in Authority Health was conducted by two researchers and documented the case study in two phases, before and during COVID-19 (May to December 2020).

Table 2. Continued.

Expanding community outreach	Develop relationships with partners to reach patient populations	Support education to those who attend churches, work in the areas, and homeless population	Church, employer and DRP and others	Leverage the partnerships to create new services

PHASE 1: PRE-COVID-19 (JANUARY TO MARCH 2020)

One of the researchers, in addition to the director of medical education, spent seven days documenting the current process and potential for telehealth support for patients at home, even prior to COVID-19, to make the services convenient to patients at home. The study material was used, along with organizational structure and mission/vision/strategy documents to submit to an NIH grant.

PHASE II: DURING COVID-19 (MAY TO DECEMBER 2020)

One-hour interviews with the director of medical education were audiotaped in the early phases of COVID-19, and two times since, to understand the COVID-19 challenges as a part of population health research. The documentation regarding the changes made during COVID-19 (as documented in the case discussed here) was reviewed and evaluated by the director of medical education.

Table 1. Data Collected.

Source	Number	Length
Annual reports (2 years; 2 people)	4	606 pages
Semi-structured interviews and written field notes from the respondents of division director, department manager, and assistant manager.	3	60 minutes-4 hours
TSMC Case Study	3	114 pages

Source	Date
Annual reports (2 years; 2 people)	2019-2020
Semi-structured interviews and written field notes from the respondents of division director, department manager, and assistant manager.	October 2020
TSMC Case Study	2002, 2006, 2009

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Caption: Figure 1. Value Lens: Customer and Business Value Cycles.

Caption: Figure 2. Key Activities to Build and Sustain Value Cycles

Caption: Figure 3. Key insights on organizational capabilities

Table 1. Summary of activities of three organizations before and during COVID-19.

Key Insights	Engagement and Enablement of Value Co-creation	
	Activities Pre-COVID-19	Activities during COVID-19
LEGO	Design a special website for a fan group Provide meaningfulness to the product development Generate competence in fan belief of value creation	Develop the activity #letsbuildtogether to share building ideas Develop a new Sesame Street special episode to support and address children's feelings about COVID-19
Marriott	Align its reward program to empower enthusiastic players Participate and share special events of customers	Implement touchless services where guests can check in, access their rooms, make special requests, and order room services without contact Allow guests to cancel reservations up to 24 hours prior to travel
Kaiser Permanente	Develop a digital platform to allow access to healthcare data Improve self-determination to change work behaviors	Implement "virtual first" strategy for non-COVID-19 patient care Use digital platform to access COVID-19 information proactively Waive costs for member patients receiving COVID-19 treatment

Key Insights	Extended services and Elicitation of Partners	
	Activities Pre-COVID-19	Activities during COVID-19

LEGO	Collaborate with young customers and create competition among teams Build selected products for commercial-ization. Reward project creator	Extend partnership with Tencent to offer more creative and safe digital play experiences for children Partner with UNICEF to support parents and communities to provide care and playful learning opportunities Partner with Pinterest to create activity-related content to support those stuck at home during COVID-19
Marriott	Partner with geo-fencing technology provider Collaborate with customers to enrich their overall experience on its property	Develop distinct partners to provide services that allow touchless disinfecting capabilities Offer food delivery services in many international locations
Kaiser Permanente	Partner with technology vendors to build portals used by customers for self-management of their health	Partner with skilled nursing facilities, residential care facilities, and families at home to provide care with remote monitoring Gather data from EHR and biometric information in each locale and share it with public health officials Partner with National Health Care for the Homeless Council to address the needs of homeless populations Partner with CDC and ten other

organizations to develop evidence based best practices to strengthen the public health infrastructure to counter COVID-19 Partner with Dignity Health, State of California, and Los Angeles County to create additional beds and ICU capacity

Table 2. Summary of activities of authority health before and during COVID-19.

	Pre-Covid	Under COVID-19
Equipment for the staff	Not applicable	Work with physicians in health systems to acquire PPE, gowns, and handwashing, etc.
Testing staff and patients	Not applicable	Set up testing centers and use local testing labs
Cleanliness and social separation	Standard cleaning and disinfecting protocols	Modify cleaning procedures, staggered work schedules, and set employee expectations
Patient access to staff	In-person access at primary care center	Use a hybrid model of telehealth and face to face visits
Staffing resource flexibility	Not applicable	Use virtual interaction and telehealth to facilitate meetings and support those who are sick by having them stay at home
Education to stakeholders	Procedures to interact with partners with roles and responsibilities	Create education material to address COVID-19 info and protocols for prevention
Expanding community outreach	Develop relationships with partners to reach patient populations	Support education to those who attend churches, work in the areas, and homeless population
	Partners	Post-Covid
Equipment for the staff	FQHCs and U.S. and international suppliers	Use this experience for planning resources in the future
Testing staff and patients	Testing labs, and churches	Use testing experience to

	to host testing	do preventive screening of patients for different disease conditions
Cleanliness and social separation		Use these protocols to reduce future infections
Patient access to staff	Tele-health software (Doxy.me)	Potential use to continue hybrid model to expand and facilitate patients access to primary care and consultation
Staffing resource flexibility	Health systems and medical students/ residents, technology providers	Providing flexibility to staff and use virtual services to improve productivity
Education to stakeholders	Public health agencies, CDC, and training specialist to prepare new material	Leverage the experience to reach patients and communities
Expanding community outreach	Church, employer and DRP and others	Leverage the partnerships to create new services

Table 1. Data Collected.

Source	Number	Length	Date
Annual reports (2 years; 2 people)	4	606 pages	2019–2020
Semi-structured interviews and written field notes from the respondents of division director, department manager, and assistant manager.	3	60 minutes-4 hours	October 2020
TSMC Case Study	3	114 pages	2002, 2006, 2009

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